ITEC 2150 WINDOWS ADMIN 2

Name: Blong Lor_____ **Total points: 60**

Instructions: For this test, you will need to answer the following questions. You may use any resource found on D2L or Microsoft Learn website, including labs, personal notes or previous Zoom sessions to assist you. Include at least one screenshot (output) for each question and then upload to D2L Midterm Dropbox.

1. Server selection Scenario. (20 points)

Background: You and your partner are Network Support Techs at ABC Tech Corporation, which has 300 employees. Your supervisor has just informed you that together you will be managing a Server 2022 Operating System Installation project involving several servers like HP & Dell name brands. Your supervisor expects you to meet in three weeks deadline and track all tasks for the project using project management software and print weekly project status reports for the top IT managers. Your company doesn't currently have any server buying guidelines or old server purchase papers.

So, in this scenario you will need to use the below table or **Microsoft Excel Spread-Sheet,** do the following tasks and answer the following questions. Then save and post it to the D2L Midterm Dropbox.

1. Get on the internet and from the Google search engine, look for companies that sell servers. The search is likely to return many results, and it may be necessary to restrict your search to **HP, Dell**, and others (If you can't locate them, select or choose any server that you may like).
2. Select and browse the vendors' websites and locate and choose a good server from each manufacture (two servers).
3. You will need to price out the servers based on the information on the below table and each one's recent reviews and popularity. **(10 points)**

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Server Hardware	SERVER ONE	SERVER TWO	
Name of the manufacturer /Seller	HPE ProLiant DL380 Gen11 6530 Server	DELL PowerEdge R660 Rack Server	
List any CPU specifications	32 cores 2.10 GHz 160 MB L3 270W	8 cores 1.8 GHz 22.5 MB 125W	
Power Supply:	2x HPE 1000W Flex Slot Titanium Plug Power Supply Kit	1x (1+0) Non Redundant, Hot-Plug Power Supply, 800W MM (100-240VAC)	
Hard drive size and type (IDE, SATA, SCSI):	2x HPE 960GB SATA 6G Read Intensive SFF BC Multi-Vendor SSD	1x 1.2TB Hard Drive SAS 12Gbps 10k 512n 2.5in Hot Plug	
Memory type/speed/	2x 32GB HPE DDR5 Smart Memory/5600 MT/s	1x 16GB RDIMM, Single Rank/5600 MT/s	
Number of memory slots if applicable;	32 DIMM slots	16 DDR5 DIMM slots	
Additional Internal/External storage:	8 SFF internal slots	10-bay version	
Number of total expansion slots:	10	3	

External interfaces/ports:	FRONT 1x USB 3.0 (iLO service port) 1x USB 3.2 Gen1 port 1x VGA Port LED Panel BACK 2x USB 3.2 Gen1 Connectors 1x VGA Connector 1x Dedicated iLO Management Port 2x LAN Ports Serial Port UID LED	FRONT 1x USB 2.0 port 1x iDRAC Direct port 1x VGA port Slide-out label with NIC/MAC address BACK 1x Dedicated iDRAC Ethernet port 1x USB 2.0 port 1x USB 3.0 port 1x VGA port 2x LOM ports 3x PCIe Gen5 slots	
On-board network Cards:	None	None	
Server Operating system Edition	HPE Integrated Lights-Out 6 (iLO 6) server management software	Windows Server 2025 Datacenter w/ reassign, 16CORE,FI,No Med,UnLTD VMs, Multi Language	
Server Case format (Slim, Rack, Tower):	2U rack-mount Rack-mountable	1U rack-mount Rack-mountable	
Total Price:	$10,955.00 USD	$14,531.42 USD	

4. What are the relative strengths and weaknesses of each server? **(5 points)**

For the HPE server, some of its strengths are strong performance and computing due to its scalable processors, massive memory due to its support for up to 8TB of DDR5 memory, and modern management with its iLO 6 software providing secure remote management. Its weaknesses include the complexity in its configuration, its high cost, and its power and cooling requirements being relatively high due to its high performance.

The DELL server's strength includes its high-performance density allowing for up to 64 cores into a 1U chassis, versatile storage options that support up to 10 2.5-inch drives, and its efficient cooling that features Direct Liquid Cooling (DLC) and Smart Flow chassis. Its weaknesses include limited expansion options, storage limitations, and relatively high cost to performance.

5. What Microsoft server 2022 edition or version would you choose to install the servers that you have selected in the scenario and listed in the above table and why? **(5 points)**

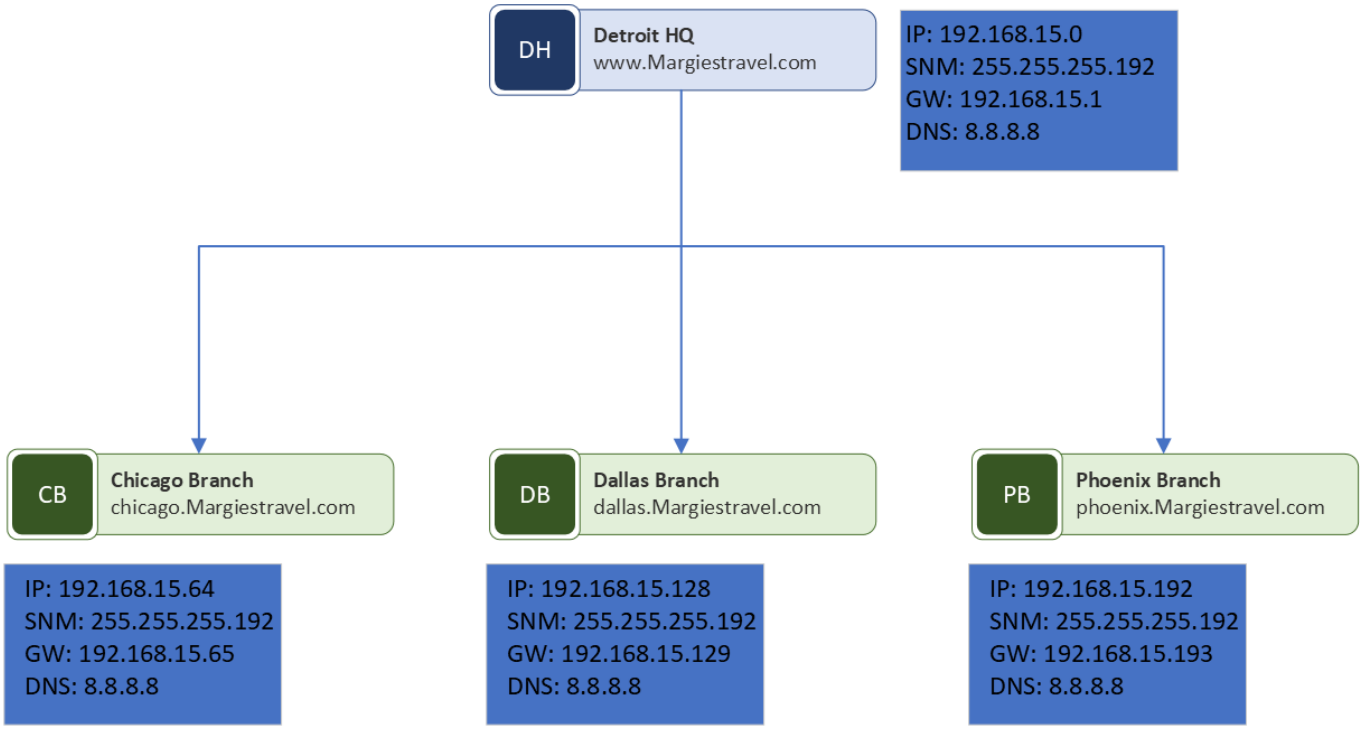
For the Dell server above, I would choose to install Windows Server 2022 Datacenter to meet the needs of the 300 employees. It would allow for unlimited Virtual Machines giving employees the ability to work and scale efficiently. For the HPE server above, I would also use the Datacenter version for the same reasons of security and scalability.

2. Scenario: Designing an Active Directory (10 points)

Margie's Travel has decided to install a Windows Server 2019/22 network. They plan to use the name **Margies*travel.com*** as their DNS name, because it is already registered with the InterNic. The corporate headquarters is located in **Detroit** and branches are located in **Chicago**, **Dallas**, and **Phoenix**. Margie's Travel wants to maintain a separate child domain in the forest representing each of the branches for ease of management.

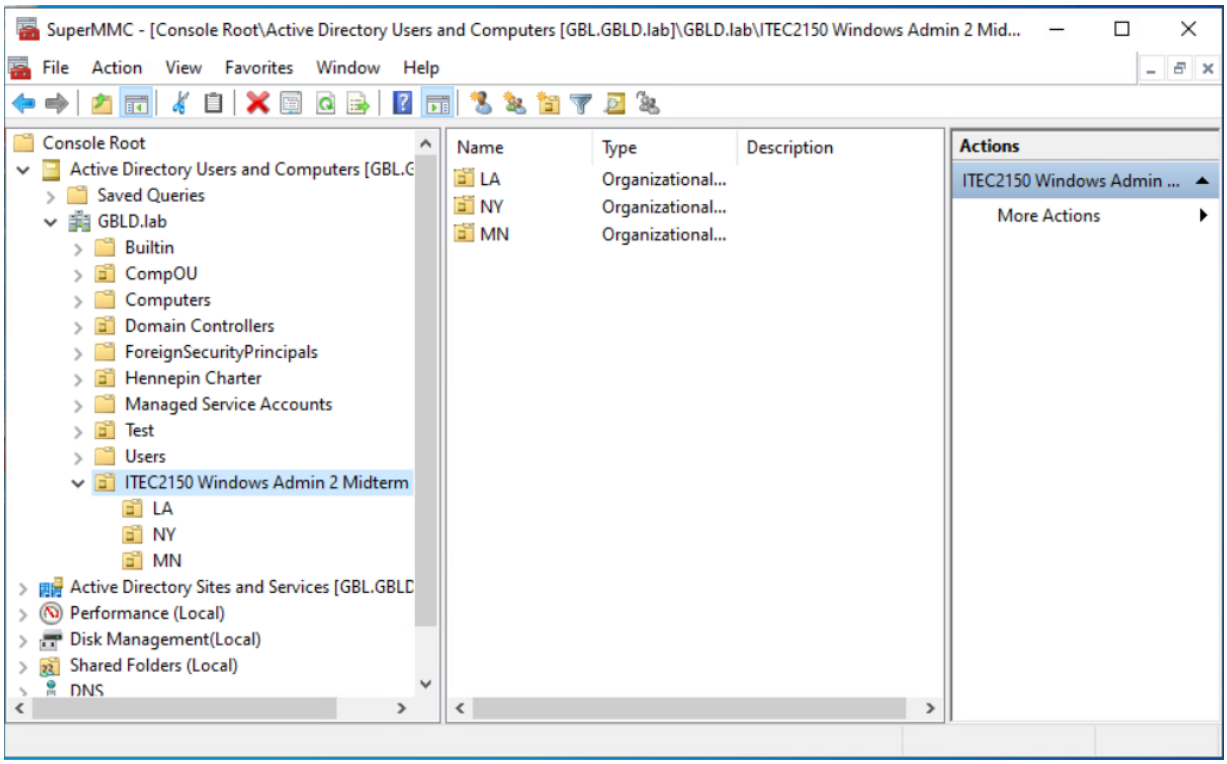
A. Create a drawing of how you would design this forest structure *(use Paint or Microsoft Visio or other drawing program).* **Be sure to include the recommended domain names within the forest. *(5 points)***

B. Create an IP address plan that will accommodate all the Margie's Travel office locations by using 192.168.15.0 /24 *(Hint: you may need four subnets minimum).* ***(5 points)***

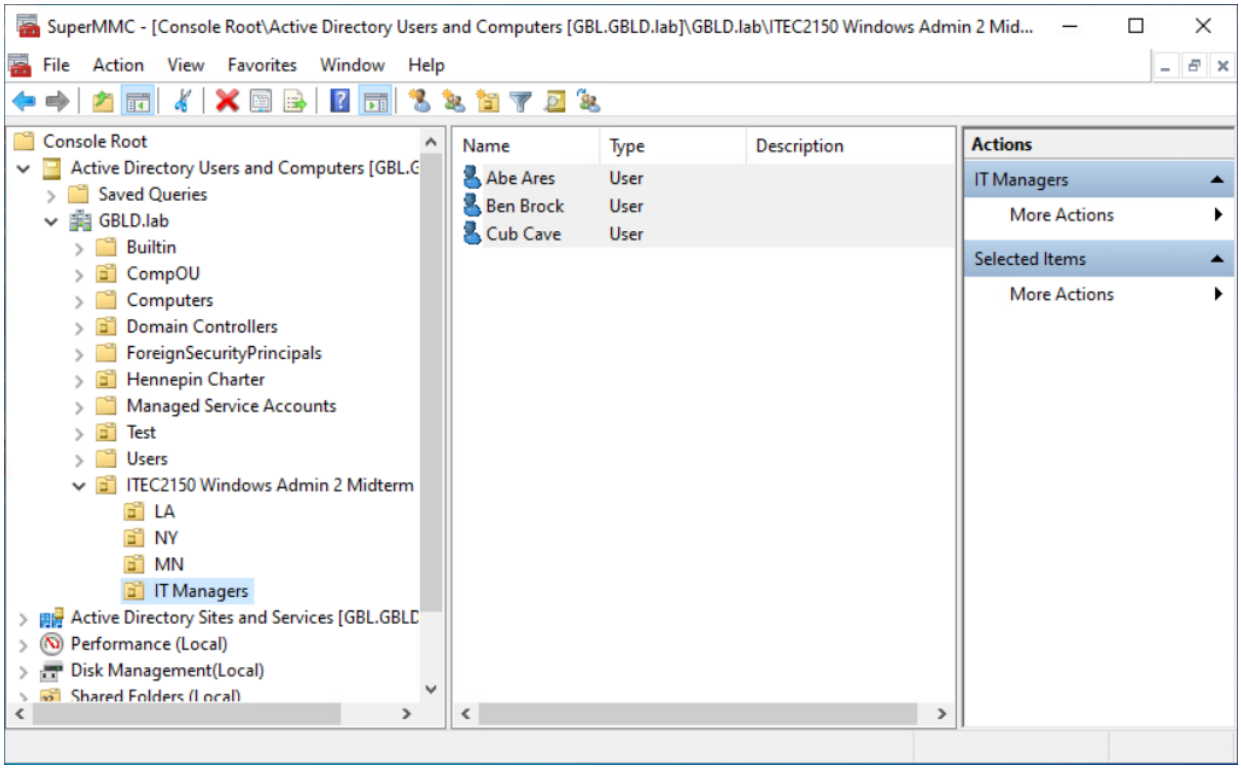


3. Complete the following tasks and submit at least one screenshot. *(30 points)*

A. Create three new OU called **LA, NY, & MN** using any tool you like. Then submit a screenshot of its output. **(10 point)**



B. Create three new usernames in a new OU called **IT Managers** using any tool you like. Name the users as you like. Then submit a screenshot of its output. **(10 points)**



C. Create three new Computer names in a new OU called **Workstations** using any tool you like and name them as **Comp1, Comp2, & Comp3.** Then submit a screenshot of its output. **(10 points)**

